Predicting Traffic Impact Severity

Machine Learning Approaches to Car Accident Data

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Executive Summary

By leveraging Random Forest models on US accident data, we can predict traffic disruption severity with **~92% accuracy** and a **0.97 ROC AUC**.

This capability allows first responders to allocate resources proactively, recognize physical lane obstructions instantly from text reports, and minimize compounding gridlock.

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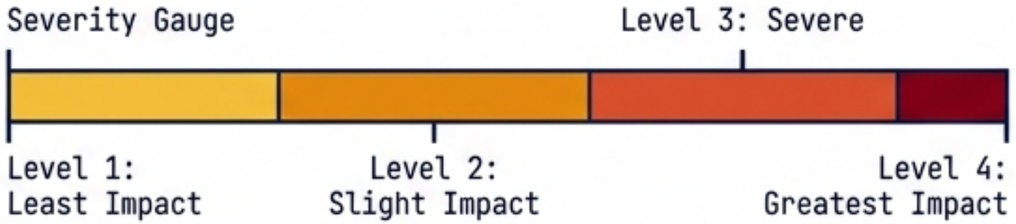
The Cost of Collisions



Collisions have a cascading ripple effect. Blocked lanes, road closures, and rubbernecking compound delays for everyone on the road.

If we can identify the specific features that make an accident traffic-severe, law enforcement can optimize emergency response times and direct extra resources to high-risk areas.

Quantifying Severity



Severe Class Imbalance

The dataset (7M+ records, 2016-2021) is highly skewed.

~79% of all accidents are Severity 2. Only ~5% are Severity 4.

Predicting the most extreme, disruptive accidents is fundamentally difficult because they are rare.

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The Data Engine: From Raw Records to Predictive Signals

Scale Reduction



Due to intense compute limits, the 7M+ record dataset was reduced to a random 10% sample.
(Irrelevant features like ID and Source were dropped).

Natural Language Processing (The Differentiator)



Bag-of-Words Encoding
The free-text "Description" column written by humans was converted into a 500-word vocabulary, allowing the model to "read" context that structured columns miss.

Overcoming Imbalance



SMOTE Balancing
Instead of just duplicating rare rows, the Synthetic Minority Oversampling Technique (SMOTE) mathematically synthesized new, realistic examples of the rare Severity 1 and 4 accidents so the models wouldn't ignore them.

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Algorithm Evaluation: Random Forest vs. KNN

Random Forest

Configuration:
Plateaued at 75 trees and 35 max features.

Accuracy:
~92% (F1 Score: ~0.92).

Dimensionality Handling:
High. Effortlessly managed the 500+ text features from the Bag-of-Words without losing signal.

Reaction to SMOTE:
Robust. Maintained high overall accuracy while successfully leveraging synthetic data to find edge cases.

K-Nearest Neighbors

Configuration:
Best performance at k=15 (features normalized).

Accuracy:
~75% (Plummeted to **~66%** when balanced).

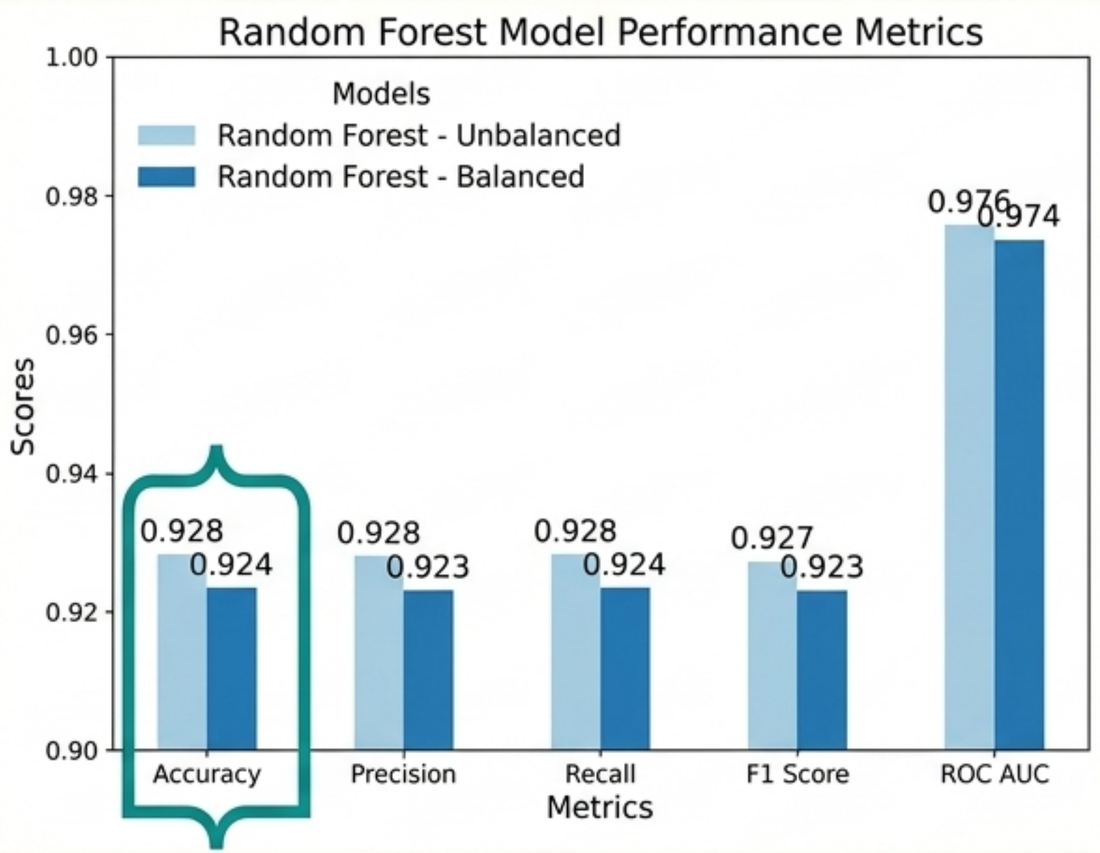
Dimensionality Handling:
Low. Suffered from the 'curse of dimensionality.' The sheer volume of 500 text features diluted the distance calculations between points.

Reaction to SMOTE:
Poor. The synthetic data points created by SMOTE introduced noise that severely skewed KNN's nearest-neighbor distance metrics.

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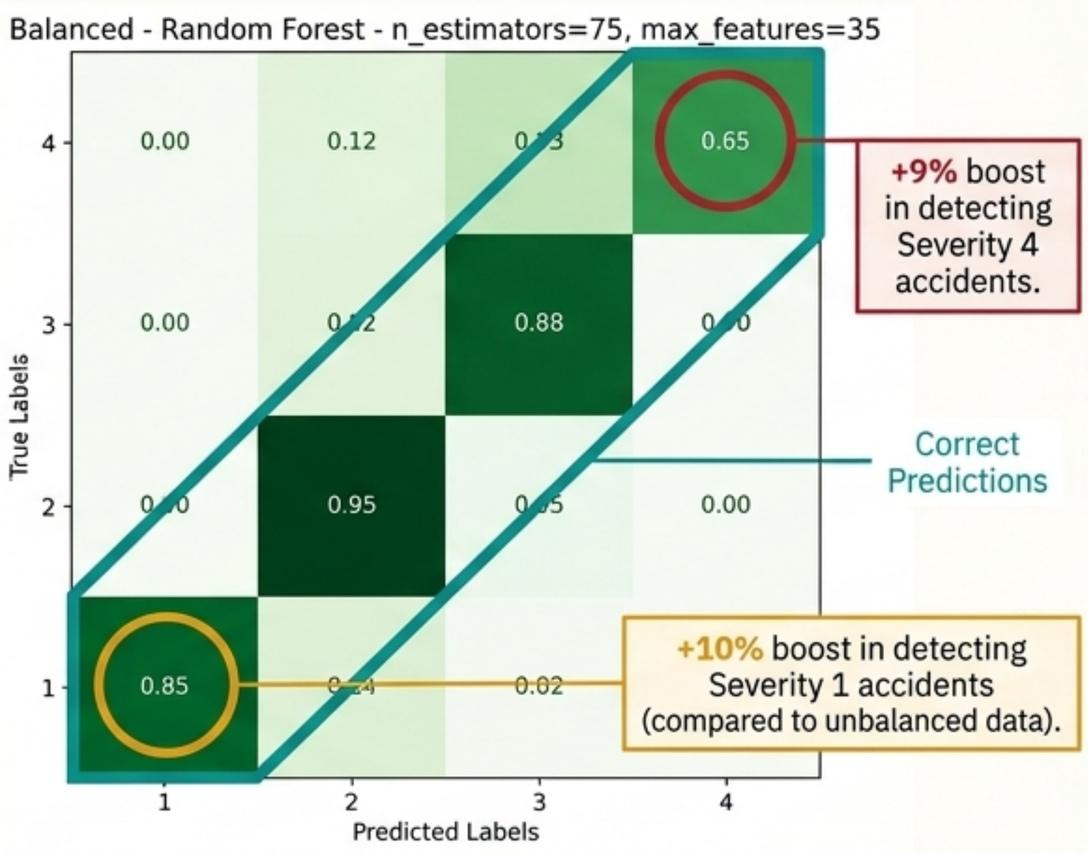
The SMOTE Trade-off: Sacrificing Micro-Fractions for Macro-Gains

The Aggregate View



Synthesizing data cost a negligible fraction of overall accuracy (dropping just 0.004 points).

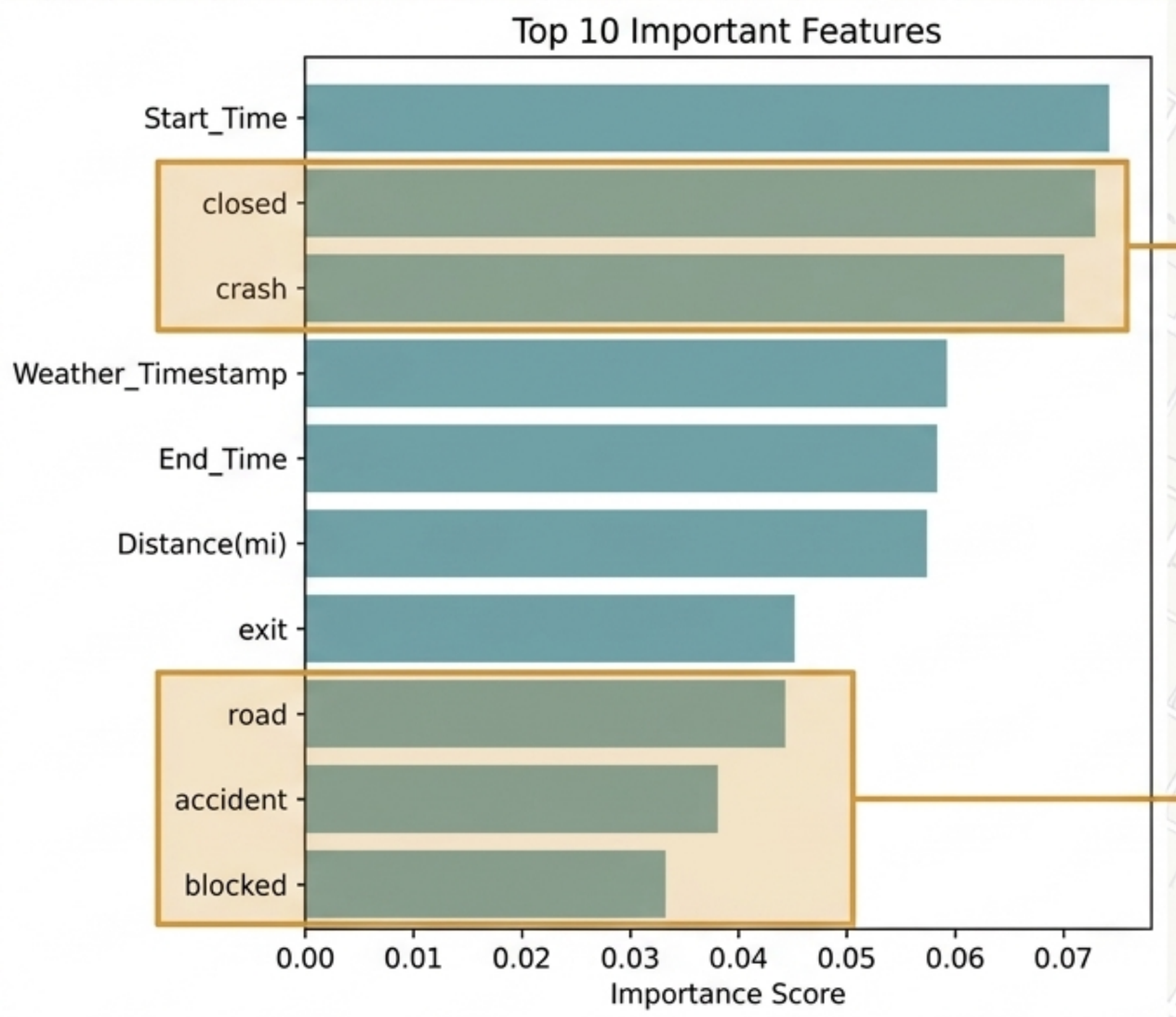
The Edge-Case Victory



By balancing the data, the model became vastly superior at identifying the rarest, most disruptive accidents—exactly what emergency dispatchers need

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What Drives Severity? The Human Element Outperforms Structured Data



Time is the Baseline

Start_Time is the single most critical feature for predicting severity. This aligns with rush-hour congestion peaks where a minor incident cascades into major delays.

Language Reveals Physical Truth

The model proved that specific words from the human-written descriptions ('closed', 'crash', 'blocked') hold massive predictive weight.
These unstructured, free-text words successfully capture real-world physical lane obstructions that standard, structured dataset columns completely miss.

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Translating Predictive Insights into Proactive Policy

Real-World Application

Predictive analyses of collision data can proactively limit severe traffic delays. By integrating models that automatically recognize high-impact features (like specific accident start times and text indicating lane closures), policymakers and dispatchers can optimize emergency responses to keep roadways clear before gridlock sets in.

Hardware Limits & Future Work



- **Current Constraints:** The present models were bottlenecked by compute limits, restricting training to a 10% data sample and a rigid 500-word NLP vocabulary.
- **The Next Frontier:** Upgraded hardware would allow processing of the full 7M+ dataset. Furthermore, integrating richer text representations (like TF-IDF or deep embedding-based text features) would capture even deeper nuances in accident reports, unlocking even higher precision.

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